

Nishant Gadde

nishantg@utexas.edu | linkedin.com/in/nishant-gadde | github.com/Nishant27-2006

EDUCATION

The University of Texas at Austin

Expected May 2028

B.S. in Electrical and Computer Engineering (Engineering Honors Program) and Mathematics

GPA: 3.87/4.0

- **Coursework:** Data Structures & Algorithms, Embedded Systems, Linear Signals & Systems, Digital Signal Processing, Circuit Theory, Linear Algebra, Differential Equations, Discrete Math, Probability & Random Processes, Theory of a Complex Variable

EXPERIENCE & RESEARCH

Machine Learning Engineer Intern, AI Infrastructure

Jun 2026 – Aug 2026

Apple

Seattle, WA

- Worked a graduate-level AI infrastructure role as an undergraduate to support serving for 50+ production models.
- Optimized low-latency inference serving via request batching and caching, boosting GPU accelerator utilization by 40%.
- Built model deployment tooling to maintain 99.99% serving reliability in an environment handling over 10,000 queries-per-second.

Research Assistant, Vision-Language Models

Jan 2026 – Present

Oden Institute for Computational Engineering & Sciences, UT Austin

Austin, TX

- Engineered A4D functional-latent-space framework for VLMs, achieving 94% inference accuracy, surpassing GPT-5 by 15%.
- Optimized unseen-affordance learning pipelines to cut new-concept acquisition costs by 10×, raising accuracy from 70% to >90%.
- Deployed isotonic-calibrated uncertainty fallback to reduce inference latency to 22 ms (100× faster) at 93% accuracy.

Machine Learning Engineer Intern

Jun 2025 – Aug 2025

Rayfield Systems

Houston, TX

- Engineered PyTorch LSTM models for high-frequency sensor forecasting that outperformed regression industry baselines by 30%.
- Deployed Spark/Kafka streaming inference with weight quantization, reducing edge memory footprint by 4× for energy companies.
- Architected full-stack Next.js web applications to deploy these predictive models used by 10+ clients for commercial use.

Research Assistant

Jun 2024 – Aug 2025

Stanford Sustainability Systems Laboratory (S3L)

Remote

- Co-authored IEEE CAI paper approximating AC optimal power flow, achieving <1% optimality gaps at 1000× faster solve times.
- Enforced operational feasibility via SOCP convex relaxations during training and applied PCA grid feature compression.
- Benchmarked framework performance against 5 IEEE systems, demonstrating convergence on GB-reduced transmission network.

SELECTED PUBLICATIONS

Siva, R.*, Bhatt, N. P.*, Yang, Y., Lee, S., **Gadde, N.**, Ellis, C., Velasquez, A., Wang, Z., & Topcu, U. “What Objects Enable, Not What They Are: Functional Latent Spaces for Affordance Reasoning,” *arXiv*, 2026.

Miltner, M., Vařata, D., Friedjungová, M., **Gadde, N.**, Štogl, O., Stary, O., & Rajagopal, R. “Towards Real-Time Machine Learning Approximations of AC Optimal Power Flow,” *2025 IEEE Conference on Artificial Intelligence (CAI)*, 2025.

Gadde, N., Mohapatra, A., Kanaparti, R., Venkatesh, A., & Manukonda, S. “Automated Detection of Heart Disease Using PhysioNet ECG Signals and Machine Learning Models,” *2024 5th International Conference on Machine Learning and Computer Application (ICMLCA)*, 2024.

PROJECTS

High-Throughput ITCH Market Data Feed Handler | C++20, UDP Multicast, SIMD, Linux socket API

2026

- Engineered a zero-allocation NASDAQ ITCH 5.0 feed handler in C++20, processing raw UDP multicast packets for real-time limit order book reconstruction.
- Implemented AVX-512 SIMD binary message parsing and custom memory pooling, reducing wire-to-book update latency to under 800 nanoseconds.
- Designed a lock-free ring buffer processing pipeline with sequence tracking and automatic gap-fill recovery, sustaining 5M+ messages per second.

Ultra-Low Latency Trading Infrastructure Library | C++17, POSIX, lock-free concurrency, kernel tuning

2025

- Architected a custom memory management subsystem featuring thread-local memory pools to eliminate OS heap allocation overhead in the critical trading path.
- Engineered a cache-aligned, single-producer single-consumer lock-free queue, preventing false sharing and minimizing inter-thread latency to 15 nanoseconds.
- Built Linux-specific system wrappers for thread CPU affinity pinning and hardware timestamping via the rdtsc instruction for precise performance profiling.

HONORS & COMPETITIONS

USACO Platinum Division, USA Computing Olympiad

2024–2025

- Qualified for the Platinum division (USACO’s **top tier**, ~**top 1%**), solving advanced graph, DP, and data-structure problems.
- Ranked **top 120 worldwide** in the 2024–25 competitive-programming season (December - US Open).

TECHNICAL SKILLS

Programming: C++ (*primary*), Python, SQL, Java, Rust, MATLAB, Bash

Math & Quant: Probability & Statistics, Linear Algebra, Stochastic Calculus, Optimization, Numerical Methods

ML & Systems: PyTorch, TensorFlow, XGBoost, Spark, Kafka, Docker, Linux, Git, CMake

Quant Methods: Market Microstructure, Time-Series Analysis, Backtesting, Monte Carlo, Kalman Filtering, Cointegration

Finance: Options & Derivatives Pricing, Greeks, PnL Attribution, Risk Metrics (VaR, Sharpe), Mean Reversion

Libraries: NumPy, pandas, SciPy, statsmodels, scikit-learn